The bowler does his run-up; you see an information titbit next to the bowler that states his name. When he delivers his ball, you see the speed in kmph. You're *seeing all this with your eyes*, because your goggles are tuned in to various information sources—and they are displaying the "extra" information, the primary information being the plain scene of the ground. (Everything *is* information, right?) *That's* AR.

Then, look at the scene below. If you haven't been to our Fair City, you might not know the monument on the left is The Gateway Of India—so your goggles tell you that bit. You can also see whether rooms are available at the Taj just opposite, and more... That's the vision some have for AR. Well, not *the* vision; it's an example application that can aid tourists. The actual number of applications for AR are very, very many.

An aside: we've overdone the goggles. AR can also be done using Head-Mounted Displays—the way it's being done in most research labs now. It could be a cell phone with a camera. It could even be implants in your eyes.

How?

If you've been brought up normally, you're probably wondering how AR can possibly work. Where does the extra information come from? How can it be overlaid? In fact, those happen to be two core questions. The first is to get the information; the second is to overlay it so perfectly on what your eyes see that it becomes one blended scene. Yes, there can be augmented hearing and smell and all, but let's keep it simple and only talk about augmented vision.

A lot depends on the complexity of what is to be achieved. Think of a relatively simple application, like helping a newbie repair computer hardware. When he looks at a motherboard through his AR system, it could label the ports, slots, jumpers, and such. What this demands is an

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information system that knows the motherboard in its entirety, well enough to label it. Why it's relatively simple is because the system need not know where the person is.

Extend this to the picture on the previous page. What needs to be known? Obviously, your physical location, and a database of things at that location. It's easy to imagine the former being supplied by a GPS device. (Easy to imagine, but hard to achieve—imagine GPS to the accuracy of a millimetre!) It's also easy to imagine the latter being supplied by a database created by a tourism company. An ad agency could supply additional information to that database—that's how you'd get the price of the car and where it's available.

Things could get more advanced: picture someone walking down the street wearing Guess jeans, and an ad popping up telling you what store is discounting them! Now that does sound far out, but it's not impossible. Think, for instance, of a world where everything had RFID tags on them.

What is called "registration," however, is a rather hard problem. It's about the extra information being properly overlaid on the real scene. When you move your eyes to the left, the tags need to shift appropriately so they still point to the correct objects—and that requires a complex head-tracking mechanism. We cannot here go into the details of how that is achieved; in any case, we're trying here to tickle your imagination, not go into boring technical details!

AR At Work

Although AR has not been entirely ignored, science fiction writers tend to emphasise the VR experience. It's cooler. But AR is as much of an exciting possibility, if not more so, than VR. With an appropriate technique, AR can enhance any scene. Scour the Web and you'll see what countless people have envisioned. Here's a little sampler.

- A mechanical engineer is working with a piece of machinery he's never touched before. As he wields his gas cutter, he sees, neatly labelled, the parts of the object he's not familiar with.
- A doctor needs to make a cut in an open brain, and his AR system shows him precisely where it is safe to make the cut.
- A road is being dug up, and water pipes are to be avoided in the process. Say there's a database of the pipes in the city. You can then imagine a camera-based AR system that allows the diggers to "look" at the pipes under the road.

We haven't even begun to scratch the surface. Those examples were just to get *you* thinking.

Who's Doing What

At the University of Cambridge Department of Engineering, they're developing what they call "Robust Model-based Tracking for Outdoor Augmented Reality." You point the AR device at an object, and the device detects the edges. Any labelling—as in augmenting—can be done only after the edges have been determined.

Edge detection—seem simple? Actually, the technicalities are way too heavy to go into here! Even for this starting point—detecting edges—you need gyroscope measurements, measurement of